

Independent Learning of Nash Equilibria in Partially Observable Markov Potential Games with Decoupled Dynamics

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Introduction

Motivation:

- Markov games model multi-agent reinforcement learning, and typically assume full access to state information.
- In many cases state is noisy or partially observed: autonomous driving, multi-robot control, strategic games of imperfect information.

Open problem:

Learning in partially observable Markov games (POMGs) with provable convergence to Nash equilibria without relying on centralized control / information sharing, and without suffering from the curse of multi-agency.

Contributions: We study POMGs with *decoupled dynamics* and *potential structure*. Extending recent advances in finite-window methods for POMDPs to the game setting, we approximate a POMG $\mathcal{P}$ by a Markov game $\mathcal{G}^m$, and show:

- Approximate **Nash equilibria** (NE) of the Markov game $\mathcal{G}^m$ are approximate NE of the original POMG (see Proposition 1).
- The game $\mathcal{G}^m$ has a **near-potential structure** (see Proposition 2).
- Our Algorithm 1 provides an **independent learning method** with convergence to NE in such POMGs. We obtain quasi-polynomial computational and sample complexity, and **no exponential dependence on N** , the number of players (see Theorem below).

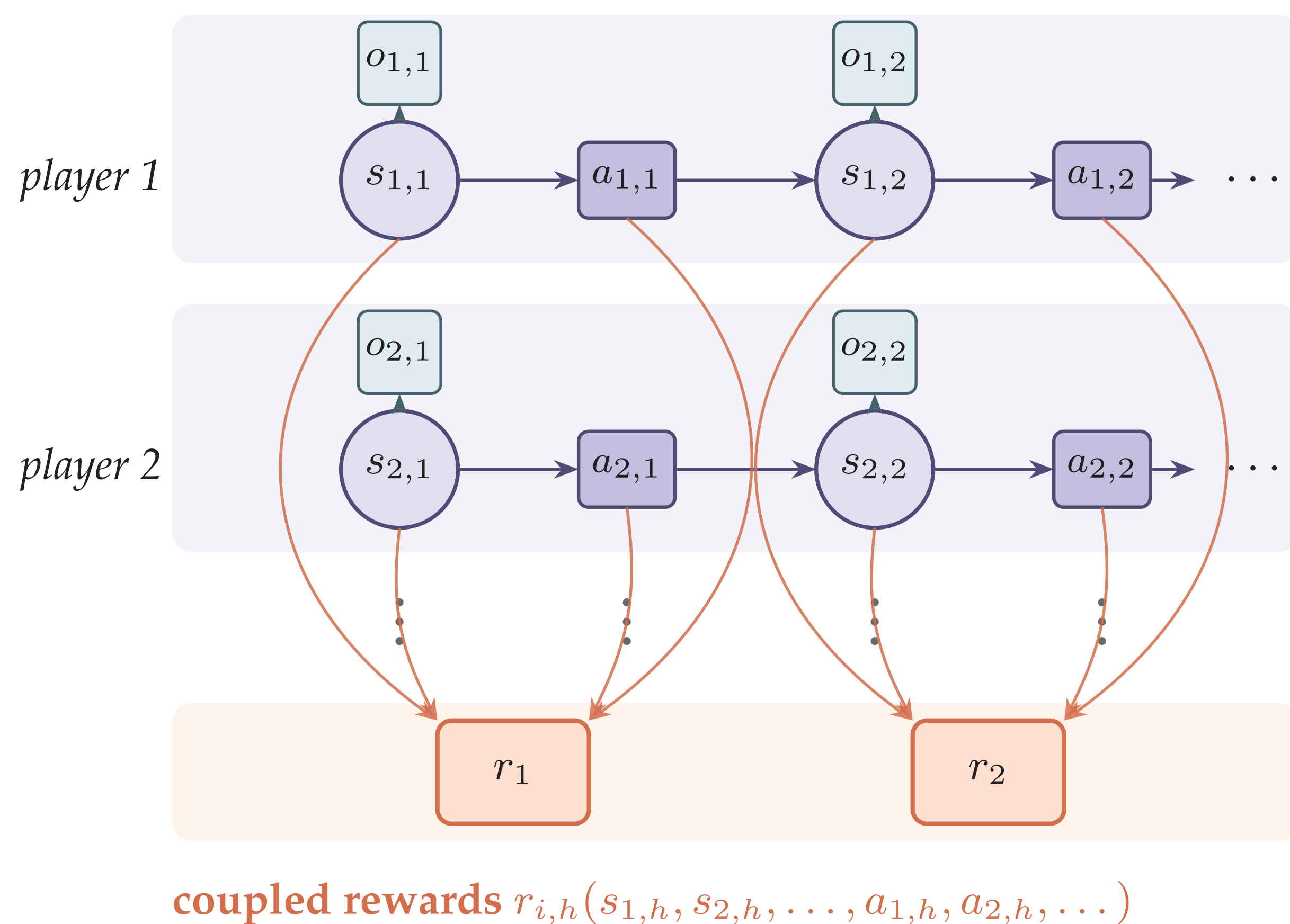
Problem Setting

Decoupled partially observable Markov game (POMG).

- players $\mathcal{N} := \{1, \dots, N\}$, finite horizon H , and tabular setting
- state and action spaces $\mathcal{S} := \mathcal{S}_1 \times \dots \times \mathcal{S}_N$ and $\mathcal{A} := \mathcal{A}_1 \times \dots \times \mathcal{A}_N$
- state is unobserved, observation space is $\mathcal{O} = \mathcal{O}_1 \times \dots \times \mathcal{O}_N$

decoupled local dynamics $\mathbb{P}_{i,h}(s_{i,h+1} \mid s_{i,h}, a_{i,h})$

decoupled local observations $\mathbb{O}_{i,h}(o_{i,h} \mid s_{i,h})$



coupled rewards $r_{i,h}(s_{1,h}, s_{2,h}, \dots, a_{1,h}, a_{2,h}, \dots)$

Histories, policies, and value function.

- player i 's h -step action-observation **histories** $\mathcal{H}_i^h := (\mathcal{A}_i \times \mathcal{O}_i)^h$
- for each player $i \in \mathcal{N}$, class of **history-dependent policies** is $\Pi_i^H := \{\pi_i = (\pi_{i,h})_{h \in [H]} \mid \pi_{i,h} : \mathcal{H}_i^{h-1} \rightarrow \Delta(\mathcal{A}_i) \text{ for all } h \in [H]\}$,
- given history $\tau = (\tau_1, \dots, \tau_N) \in \mathcal{H}^{h-1}$, $a = (a_1, \dots, a_N) \in \mathcal{A}$ is chosen by $\pi = (\pi_1, \dots, \pi_N) \in \Pi_1^H \times \dots \times \Pi_N^H$ with probability $\pi_h(a \mid \tau) = \prod_{i \in \mathcal{N}} \pi_{i,h}(a_i \mid \tau_i)$
- value functions:** for $\pi \in \Pi^H$, let $V_i(\pi) := \mathbb{E}_\pi[\sum_{h=1}^H r_{i,h}(s_h, a_h)]$

Learning objective.

- for $\epsilon > 0$, we aim to find $\pi \in \Pi^H$ that is an **ϵ -approximate Nash equilibrium**, i.e., such that $\forall i \in \mathcal{N}$ and $\forall \pi'_i \in \Pi_i^H$, we have $V_i(\pi) \geq V_i(\pi'_i, \pi_{-i}) - \epsilon$

→ intractable without further assumptions on game structure and observations

Assumptions.

- potential structure:** let $\Pi_i := \{\{\pi_{i,h} : \mathcal{S} \rightarrow \Delta(\mathcal{A}_i)\}_{h=1}^H\}$, let $\Pi := \Pi_1 \times \dots \times \Pi_N$. There exists a potential function $\Phi : \Pi \rightarrow \mathbb{R}$ s.t. $\forall i \in \mathcal{N}$, $\forall \pi \in \Pi$, and $\forall \pi'_i \in \Pi_i$,

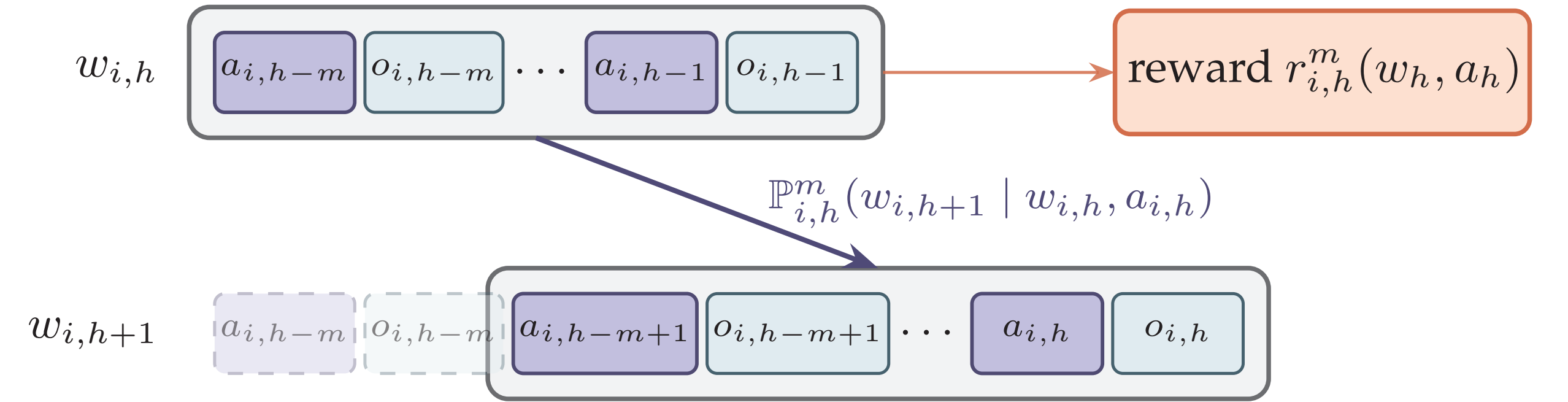
$$V_i(\pi) - V_i(\pi'_i, \pi_{-i}) = \Phi(\pi) - \Phi(\pi'_i, \pi_{-i}).$$

- filter stability:** $\exists \rho > 0$ s.t. for all $\tau, \tau' \in \mathcal{H}^h$, $a \in \mathcal{A}$, and $o \in \mathcal{O}$,

$$\|b(\cdot \mid \tau \circ (a, o)) - b(\cdot \mid \tau' \circ (a, o))\|_{TV} \leq (1 - \rho) \|b(\cdot \mid \tau) - b(\cdot \mid \tau')\|_{TV}.$$

Approximating POMGs via Near-Potential Games

Superstate Markov game $\mathcal{G}^m$. We approximate the POMG by a Markov game over m -step windows of local histories; for each player $i \in \mathcal{N}$:



Proposition 1 (Nash equilibrium approximation). Let $\epsilon > 0$, and let $\pi \in \Pi^m$ be an ϵ -approximate NE of the superstate Markov game $\mathcal{G}^m$. Under filter stability, π is an $(\epsilon + 2\epsilon_\rho^m)$ -approximate NE of $\mathcal{P}$, where $\epsilon_\rho^m := 4H^2(1 - \rho)^m$.

Proposition 2 (Near-potential). Under filter stability, there exists a function $\Psi : \Pi^m \rightarrow \mathbb{R}$ such that for all $\pi \in \Pi^m$, $i \in \mathcal{N}$, and $\pi'_i \in \Pi_i^m$, we have

$$|(V_i^m(\pi) - V_i^m(\pi'_i, \pi_{-i})) - (\Psi(\pi) - \Psi(\pi'_i, \pi_{-i}))| \leq 2\epsilon_\rho^m.$$

Independent Nash Equilibrium Learning Algorithm

- Objective:** given $\epsilon > 0$, learn a finite-window policy $\pi \in \Pi^m$ that is an ϵ -approximate Nash equilibrium w.r.t. deviations $\pi'_i \in \Pi_i^H$
- Independent learning:** players see only own actions, observations, rewards, and policies; simultaneous local policy updates (a.k.a. self-play)

Algorithm idea.

- fixing π_{-i} in $\mathcal{G}^m$ induces MDP $\mathcal{M}^{\pi_{-i}}$ averaged over (w_{-i}, a_{-i}) -visitation
- at each iteration, we estimate Q -functions of $\mathcal{M}^{\pi_{-i}}$ for each $i \in \mathcal{N}$ based on trajectory samples, and perform simultaneous (soft) policy iteration updates

Algorithm 1: Learning Finite-Window Policy in POMG (at player i)

- Initialize $\pi_{i,h}^{(0)} = 1/|\mathcal{A}_i|$ for all $h \in [H]$.
- for** $k = 0, 1, \dots, K - 1$ **do**
- Collect trajectory $\tau_i = ((a_{i,h}^{(t)}, o_{i,h}^{(t)}, r_{i,h}^{(t)})_{h=1}^H)_{t=1}^T$ by following $\pi_i^{(k)}$.
- Estimate $\hat{\mathbb{P}}_{i,h}^m$ and $\hat{r}_{i,h}^{m, \pi_{-i}^{(k)}}$ based on empirical frequencies.
- Let $\hat{Q}_{i,H+1}^{(k)}(w_i, a_i) := 0$ for all $w_i \in \mathcal{H}_i^{\leq m}$, $a_i \in \mathcal{A}_i$.
- for** $h = H, \dots, 1$ **do**
- for** $w_i \in \mathcal{H}_i^{\leq m}$, $a_i \in \mathcal{A}_i$ **do**
- $\hat{Q}_{i,h}^{(k)}(w_i, a_i) := \hat{r}_{i,h}^{m, \pi_{-i}^{(k)}}(w_i, a_i) + \sum_{w'_i, a'_i} \hat{\mathbb{P}}_{i,h}^m(w'_i \mid w_i, a_i) \pi_{i,h+1}^{(k)}(a'_i \mid w'_i) \hat{Q}_{i,h+1}^{(k)}(w'_i, a'_i).$
- $\pi_{i,h}^{(k+1)}(a_i \mid w_i) := (1 - \eta^{(k)}) \pi_{i,h}^{(k)}(a_i \mid w_i) + \eta^{(k)} \mathbf{1}\{a_i \in \arg \max_{a'_i} \hat{Q}_{i,h}^{(k)}(w_i, a'_i)\}.$

Convergence Result

Assumptions.

- sufficient **observation noise** (see Assumption 3 in [JK26])
- sufficient **exploration** of underlying states $s_i \in \mathcal{S}_i$ (see Assumption 4 in [JK26])

Theorem. Let $\epsilon > 0$, $\delta > 0$, suppose each player $i \in \mathcal{N}$ runs Algorithm 1 simultaneously with stepsizes $\eta^{(k)} = \Theta(1/\sqrt{N^2 H^3 k})$, and suitably chosen m , T , and K . Let $A := \max_{i \in \mathcal{N}} |\mathcal{A}_i|$, $O := \max_{i \in \mathcal{N}} |\mathcal{O}_i|$. With probability $1 - \delta$, $\exists k \in [K]$ such that $\pi^{(k)}$ is an ϵ -approximate Nash equilibrium of the POMG $\mathcal{P}$. Sample and computational complexity of Algorithm 1 are at most

$$\mathcal{O}\left(\left(\frac{NHAO}{\beta\alpha\epsilon}\right)^{\mathcal{O}(\rho^{-1} \log(A/(\beta\epsilon)))} \log(1/\delta)\right).$$

Conclusion & Outlook

- The **superstate Markov game abstraction** provides a useful framework for designing and analyzing principled algorithms for POMGs.
- We give a **tractable independent learning algorithm** for a relevant subclass of POMGs with convergence and sample complexity guarantee.

Open questions: where is the boundary between tractable and intractable POMG settings? Is state decoupling a necessary restriction?